

Problem 9

Problem 9

Problem 1 A piecewise defined function f is given by

$$f(x) = \begin{cases} \frac{2x-3}{x-2} & \text{if } x < 2 \\ \frac{x^2-5x+6}{x^2-4} & \text{if } x > 2 \end{cases}$$

- (a) Find all vertical asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.
- (b) Find all horizontal asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.